

Conditional Hopfield model

Setting

Consider a Hopfield model with N neurons and $P = \alpha N$ patterns $\xi^\mu \in \{\pm 1\}^N$ ($\mu = 1, \dots, P$). The patterns are random and uncorrelated, with i.i.d. entries drawn uniformly from $\{\pm 1\}$. The Boltzmann distribution reads

$$p(\mathbf{x}) \propto e^{-\beta E(\mathbf{x})}, \quad E(\mathbf{x}) = -\frac{1}{2} \sum_{i,j} J_{ij} x_i x_j, \quad J_{ij} = \frac{1}{N} \sum_{\mu=1}^P \xi_i^\mu \xi_j^\mu. \quad (1)$$

(The $i = j$ diagonal gives an additive constant $-\alpha\beta N/2$ that does not affect the physics; keeping the full sum makes the Hamiltonian manifestly $-\beta E = (\beta/(2N)) \sum_\mu A_\mu^2$ with $A_\mu = \sum_i \xi_i^\mu x_i$.)

We clamp a subset $U \subset [N]$ of the spins to the first pattern, i.e. $x_i = \xi_i^1$ for all $i \in U$. Write $|U| = (1-t)N$, so that $t = 1 - |U|/N$ is the fraction of *free* spins, and let $F = [N] \setminus U$ with $|F| = tN$. The set U is drawn uniformly at random among subsets of size $(1-t)N$; by symmetry the replica calculation does not depend on which particular subset has been chosen.

The Hamiltonian of the clamped system takes the form:

$$E_{U(x_F)} = -\frac{1}{2} \sum_{i,j \in F} J_{ij} x_i x_j - \sum_{i \in F} h_i^U x_i + \text{const}, \quad (2)$$

where the effective field on free sites is

$$h_i^U = \sum_{j \in U} J_{ij} \xi_j^1 = (1-t) \xi_i^1 + \frac{1}{N} \sum_{\mu \geq 2} \xi_i^\mu \sum_{j \in U} \xi_j^1 \xi_j^\mu. \quad (3)$$

We are interested in the retrieval properties of the first pattern as a function of (α, t) : how much information about ξ^1 do the free spins on F carry when a fraction $1-t$ of them is already pinned to ξ^1 ?

Replica computation

The disorder-averaged free energy per spin is

$$-\beta f = \lim_{N \rightarrow \infty} \frac{1}{N} \overline{\ln Z}, \quad Z = \sum_{\substack{\mathbf{x} \in \{\pm 1\}^N \\ x_i = \xi_i^1 \forall i \in U}} e^{-\beta E(\mathbf{x})}, \quad (4)$$

with overline denoting the quenched average over the patterns. We evaluate $\overline{\ln Z}$ by the replica trick,

$$\overline{\ln Z} = \lim_{n \rightarrow 0} \frac{\overline{Z^n} - 1}{n}, \quad (5)$$

introducing n copies of the system with replica index $a = 1, \dots, n$ and evaluating $\overline{Z^n}$ for integer n before analytically continuing.

Gauge transformation

It is convenient to remove pattern 1 from the disorder by performing the gauge transformation

$$\sigma_i^a = \xi_i^1 x_i^a, \quad (6)$$

which maps $x_i^a \in \{\pm 1\}$ to $\sigma_i^a \in \{\pm 1\}$ with the same support. In the new variables the clamped spins become $\sigma_i = 1$ for all $i \in U$ (since $\xi_i^1 \cdot \xi_i^1 = 1$), while on the free sites $i \in F$ the variables σ_i^a remain free binary spins. Defining the rotated noise patterns

$$\tilde{\xi}_i^\mu = \xi_i^1 \xi_i^\mu \quad (\mu \geq 2), \quad (7)$$

one has $\tilde{\xi}_i^\mu \in \{\pm 1\}$ still i.i.d. uniform by symmetry of the distribution of ξ^μ . The Hebbian energy becomes

$$-\beta E = \frac{\beta}{2N} \sum_{\mu=1}^P A_\mu^2, \quad (8)$$

with

$$\begin{aligned} A_1(\sigma) &= \sum_{i=1}^N \sigma_i = (1-t)N + \sum_{i \in F} \sigma_i, \\ A_{\mu(\sigma)} &= \sum_{i \in U} \tilde{\xi}_i^\mu + \sum_{i \in F} \tilde{\xi}_i^\mu \sigma_i \quad (\mu \geq 2). \end{aligned} \quad (9)$$

Pattern 1 contributes a deterministic “condensed” term proportional to the total magnetisation $(1/N) \sum_i \sigma_i = (1-t) + tm$ (free-sector magnetisation rescaled by t , plus the clamped contribution $(1-t)$). The remaining $\alpha N - 1$ patterns contribute an i.i.d. Gaussian-like noise that will be handled by replica/HS tricks.

Order parameters

On the free sector F we introduce the intrareplica magnetisation and the Edwards–Anderson overlap

$$m^a = \frac{1}{tN} \sum_{i \in F} \sigma_i^a, \quad q^{ab} = \frac{1}{tN} \sum_{i \in F} \sigma_i^a \sigma_i^b \quad (a \neq b). \quad (10)$$

The corresponding “full” quantities that include the clamped block are

$$M^a = \frac{1}{N} \sum_i \sigma_i^a = (1-t) + tm^a, \quad (11)$$

$$\hat{Q}^{ab} = \frac{1}{N} \sum_i \sigma_i^a \sigma_i^b = \begin{cases} 1-t + tq^{ab} & a \neq b \\ 1 & a = b \end{cases} \quad (12)$$

where $\hat{Q}^{aa} = 1$ follows from $\sigma_i^2 = 1$. If we adopt the convention $q^{aa} = 1$ the two cases collapse into the single expression $\hat{Q}^{ab} = (1-t) + tq^{ab}$ for all a, b .

Condensed pattern

The contribution of pattern 1 to the replicated action is deterministic and reads

$$\frac{\beta}{2N} \sum_a A_1(\sigma^a)^2 = \frac{\beta N}{2} \sum_a (M^a)^2. \quad (13)$$

This is the “signal” term: it favours configurations with M^a aligned to the clamped block, i.e. $m^a \approx 1$.

Noise patterns ($\mu \geq 2$)

For each noise pattern $\mu \geq 2$ and replica a decouple A_μ^a by a Hubbard–Stratonovich transformation,

$$\exp\left(\frac{\beta}{2N}(A_\mu^a)^2\right) = \int \frac{dz_\mu^a}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(z_\mu^a)^2 + \sqrt{\beta/N} z_\mu^a A_\mu^a\right). \quad (14)$$

Collecting replicas and expanding A_μ^a explicitly, the argument of the exponential becomes

$$\sqrt{\beta/N} \sum_a z_\mu^a A_\mu^a = \sqrt{\beta/N} \left[\left(\sum_a z_\mu^a \right) \sum_{i \in U} \tilde{\xi}_i^\mu + \sum_{i \in F} \tilde{\xi}_i^\mu \left(\sum_a z_\mu^a \sigma_i^a \right) \right]. \quad (15)$$

Each $\tilde{\xi}_i^\mu$ is i.i.d. ± 1 , and averaging factorises over sites:

$$\overline{\exp(c\tilde{\xi})} = \cosh(c) = 1 + \frac{c^2}{2} + O(c^4) \approx \exp\left(\frac{c^2}{2}\right) \quad \text{for } c = O(1/\sqrt{N}). \quad (16)$$

With $c_i = \sqrt{\beta/N} \times (\dots)_i$ of order $1/\sqrt{N}$, this is exact in the large- N limit up to subleading corrections. Collecting over sites,

$$\overline{\exp\left(\sqrt{\beta/N} \sum_i \tilde{\xi}_i^\mu (\dots)_i\right)} \approx \exp\left(\frac{\beta}{2N} \sum_i (\dots)_i^2\right). \quad (17)$$

Using $(1/N) \sum_{i \in F} \sigma_i^a \sigma_i^b = tq^{ab}$ and $\sum_{i \in U} 1 = (1-t)N$, the squared term expands to

$$\sum_{i \in F} \left(\sum_a z_\mu^a \sigma_i^a \right)^2 + (1-t)N \left(\sum_a z_\mu^a \right)^2 = N \sum_{ab} z_\mu^a z_\mu^b \hat{Q}^{ab}, \quad (18)$$

the $(1-t)N$ term from the clamped block contributing the same overlap correction to every replica pair. The disorder-averaged HS integrand for each $\mu \geq 2$ thus reads

$$\int \prod_a \frac{dz_\mu^a}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} \sum_{ab} (\delta_{ab} - \beta \hat{Q}^{ab}) z_\mu^a z_\mu^b\right) = \det(I - \beta \hat{Q})^{-1/2}. \quad (19)$$

Raising to the power $P = \alpha N$ and taking the log gives

$$\prod_{\mu \geq 2} \int \dots = \exp\left(-\frac{\alpha N}{2} \text{Tr} \ln(I - \beta \hat{Q})\right). \quad (20)$$

Free sector trace

After noise averaging, the effective action decouples across free sites up to the global order parameters m^a, q^{ab} . Introducing conjugate fields $\hat{m}^a, \hat{q}^{ab}$ to enforce the definitions of m^a and q^{ab} (e.g. via an integral representation of the delta function), the free-site trace is

$$Z_{\text{site}} = \sum_{\{\sigma^a\}} \exp\left(\sum_a \hat{m}^a \sigma^a + \sum_{a \neq b} \hat{q}^{ab} \sigma^a \sigma^b\right), \quad (21)$$

and the action contains the terms

$$tN \sum_a \hat{m}^a m^a + tN \sum_{a \neq b} \hat{q}^{ab} q^{ab} - tN \ln Z_{\text{site}}. \quad (22)$$

At the saddle point $\hat{m}^a$ and $\hat{q}^{ab}$ will be fixed by extremisation in m^a, q^{ab} .

RS ansatz

In the Replica Symmetric (RS) ansatz we set $m^a = m, q^{ab} = q$ for $a \neq b, \hat{m}^a = \hat{m}, \hat{q}^{ab} = \hat{q}$. Then

$$\hat{Q}^{ab} = \begin{cases} 1 & a = b \\ \tilde{q} & a \neq b \end{cases}, \quad \tilde{q} := (1 - t) + tq. \quad (23)$$

As an $n \times n$ matrix $\hat{Q} = (1 - \tilde{q})I + \tilde{q}\mathbf{1}\mathbf{1}^T$, so its eigenvalues are

$$\begin{aligned} \lambda_+ &= 1 + (n - 1)\tilde{q} \quad (\text{multiplicity } 1, \text{ eigenvector } \mathbf{1}), \\ \lambda_- &= 1 - \tilde{q} \quad (\text{multiplicity } n - 1, \text{ orthogonal complement}). \end{aligned} \quad (24)$$

Hence

$$\text{Tr} \ln(I - \beta \hat{Q}) = \ln(1 - \beta \lambda_+) + (n - 1) \ln(1 - \beta \lambda_-). \quad (25)$$

Expanding in small n with $D := 1 - \beta(1 - \tilde{q})$, so that $1 - \beta \lambda_+ = D - n\beta\tilde{q}$ and $1 - \beta \lambda_- = D$:

$$\begin{aligned} \text{Tr} \ln(I - \beta \hat{Q}) &= \ln(D - n\beta\tilde{q}) + (n - 1) \ln D \\ &= n \ln D - \frac{n\beta\tilde{q}}{D} + O(n^2), \end{aligned} \quad (26)$$

so that

$$\lim_{n \rightarrow 0} \frac{1}{n} \text{Tr} [\ln(I - \beta \hat{Q})] = \ln(1 - \beta(1 - \tilde{q})) - \frac{\beta\tilde{q}}{1 - \beta(1 - \tilde{q})}. \quad (27)$$

For the free-site trace, one uses the HS identity $\exp(\hat{q} \sum_{a \neq b} \sigma^a \sigma^b) = \int Dz \exp(\sqrt{2\tilde{q}}z \sum_a \sigma^a - n\hat{q})$ and performs the sum over $\{\pm 1\}$ at each site, giving

$$\left(\frac{1}{n}\right) \ln Z_{\text{site}} \rightarrow -\hat{q} + \int Dz \ln[2 \cosh(\hat{m} + \sqrt{2\tilde{q}}z)] \quad (n \rightarrow 0). \quad (28)$$

Extremising in $\hat{m}, \hat{q}$ (and rescaling so that $\hat{m} = \beta M, 2\hat{q} = \beta^2 \alpha r$; equivalently one introduces r as an additional order parameter conjugate to the noise-overlap so that the noise kernel enters the site trace as an effective Gaussian field $\beta\sqrt{\alpha r}z$), one ends up with the compact RS free energy

$$\begin{aligned} -\beta f_{\text{RS}} &= \frac{\beta}{2} M^2 - \frac{\alpha \beta^2 t}{2} r(1 - q) \\ &\quad + \frac{\alpha}{2} \left[\frac{\beta\tilde{q}}{1 - \beta(1 - \tilde{q})} - \ln(1 - \beta(1 - \tilde{q})) \right] \\ &\quad + t \int Dz \ln[2 \cosh(\beta(M + \sqrt{\alpha r}z))], \end{aligned} \quad (29)$$

with $M = (1 - t) + tm$, $\tilde{q} = (1 - t) + tq$, and $Dz = e^{-z^2/2} dz / \sqrt{2\pi}$ the standard Gaussian measure.

Saddle-point equations

Stationarity of Equation 29 with respect to m, q, r yields

$$\begin{aligned} m &= \int Dz \tanh(\beta(M + \sqrt{\alpha r}z)), \\ q &= \int Dz \tanh^2(\beta(M + \sqrt{\alpha r}z)), \\ r &= \frac{\tilde{q}}{(1 - \beta(1 - \tilde{q}))^2}. \end{aligned} \quad (30)$$

The first equation is the self-consistent magnetisation in an effective field βM with Gaussian noise of variance $\beta^2 \alpha r$; it is the direct generalisation of the Curie–Weiss identity. The second identifies the free-sector Edwards–Anderson parameter as the second moment of the same local field. The third encodes the feedback of the free-sector overlap on the noise variance, promoted by the clamped block ($\tilde{q} \geq q$ when $t < 1$).

Limits. At $t = 1$ we have $M = m$, $\tilde{q} = q$, and Equation 30 reduces to the classical Amit–Gutfreund–Sompolinsky (AGS) RS equations. At $t = 0$ the free sector is empty, $M \equiv 1$ and $\tilde{q} \equiv 1$, the integrals collapse, and the retrieval of pattern 1 is trivially exact.

Zero-temperature analysis

The zero-temperature limit is obtained by taking $\beta \rightarrow \infty$ while keeping

$$C_0 := \beta(1 - \tilde{q}) = \beta t(1 - q) \quad (31)$$

finite. This is the standard scaling for Hopfield-like models: deep in a retrieval minimum the spins are almost always polarised, so $q \rightarrow 1$, but $1 - q$ vanishes as $1/\beta$ times a finite factor controlled by the noise. In this limit $\tanh(\beta x) \rightarrow \text{sign}(x)$ pointwise, so

$$\begin{aligned} m &\rightarrow \int Dz \, \text{sign}(M + \sqrt{\alpha r} z) = \text{erf}(M/\sqrt{2\alpha r}), \\ 1 - q &\rightarrow \int Dz \, \text{sech}^2(\beta(M + \sqrt{\alpha r} z)). \end{aligned} \quad (32)$$

The sech^2 integrand concentrates on a thin region of width $O(1/\beta)$ around the zero of its argument. Using $\int \text{sech}^2(\beta x) dx = 2/\beta$ and approximating $\text{sech}^2(\beta(M + \sqrt{\alpha r} z))$ as $(2/\beta)\delta(M + \sqrt{\alpha r} z)$,

$$1 - q \approx (2/\beta)(1/\sqrt{\alpha r})(Dz)_{z=-M/\sqrt{\alpha r}} = \frac{\sqrt{2/\pi}}{\beta\sqrt{\alpha r}} e^{-M^2/(2\alpha r)}. \quad (33)$$

Defining $y = M/\sqrt{2\alpha r}$ gives

$$m = \text{erf}(y), \quad C_0 = \frac{t\sqrt{2/\pi}e^{-y^2}}{\sqrt{\alpha r}}. \quad (34)$$

From $r = \tilde{q}/D^2 \rightarrow 1/(1 - C_0)^2$ we get $\sqrt{\alpha r} = \sqrt{\alpha}/(1 - C_0)$, and substituting in the expression for C_0 ,

$$C_0 = u(1 - C_0), \quad u := t\sqrt{2/(\pi\alpha)}e^{-y^2} \Rightarrow C_0 = \frac{u}{1 + u}. \quad (35)$$

Finally, using $\sqrt{\alpha r} = \sqrt{\alpha}(1 + u)$ and rearranging the self-consistency $M = (1 - t) + tm = (1 - t) + t \text{erf}(y)$ as $y\sqrt{2\alpha}(1 + u) = M$, and noting that $y\sqrt{2\alpha}u = t(2y/\sqrt{\pi})e^{-y^2}$, we arrive at the single scalar equation

$$y\sqrt{2\alpha} = (1 - t) + t \left[\underbrace{\text{erf}(y) - \frac{2y}{\sqrt{\pi}}e^{-y^2}}_{g(y)} \right]. \quad (36)$$

The bias $(1 - t)$ on the right-hand side is the contribution of the clamped block: it acts as a staggered “external field” on the free spins, whose strength grows as t decreases.

Existence of a solution. Since $g(0) = 0$, $g(\infty) = 1$ and $g'(y) = (4y^2/\sqrt{\pi})e^{-y^2} \geq 0$, the right-hand side of Equation 36 ranges over $[1-t, 1)$ as y ranges over $[0, \infty)$. The left-hand side, $y\sqrt{2\alpha}$, is unbounded above. Hence a solution $y_* > 0$ always exists when $t > 0$: the $y = 0$ (unmagnetised) fixed point of the classical Hopfield model is destroyed by the bias of the clamped block.

Retrieval spinodal

The retrieval branch — the largest root of Equation 36 — can annihilate with an intermediate unstable root in a tangent bifurcation. At the bifurcation both Equation 36 and its y -derivative vanish:

$$y\sqrt{2\alpha_c} = (1-t) + tg(y), \quad \sqrt{2\alpha_c} = tg'(y). \quad (37)$$

Eliminating $\sqrt{2\alpha_c}$ by substituting the second into y times the first gives

$$t[yg'(y) - g(y)] = 1-t, \quad (38)$$

i.e.

$$h(y_*) = \frac{1-t}{t}, \quad h(y) := yg'(y) - g(y) = \frac{2y}{\sqrt{\pi}}(2y^2 + 1)e^{-y^2} - \text{erf}(y). \quad (39)$$

The function h satisfies $h(0) = 0$, $h(\infty) = -1$, and $h'(y) = yg''(y) = (8y^2(1-y^2)/\sqrt{\pi})e^{-y^2}$, so it is unimodal with a single maximum at $y = 1$:

$$h_{\max} = h(1) = \frac{6}{e\sqrt{\pi}} - \text{erf}(1) \approx 0.4027. \quad (40)$$

Equation 39 admits a solution (hence a spinodal bifurcation exists) iff

$$\frac{1-t}{t} \leq h_{\max} \iff t \geq t^* = \frac{1}{1+h_{\max}} \approx 0.7129. \quad (41)$$

Given $y^* \geq 1$ on the decreasing branch of h , the spinodal capacity is

$$\alpha_c(t) = \frac{t^2}{2}[g'(y^*)]^2 = \frac{8t^2(y^*)^4}{\pi}e^{-2(y^*)^2}. \quad (42)$$

At the cusp $y^* = 1$, $\alpha_c(t^*) = 8(t^*)^2/(\pi e^2) \approx 0.1753$, which is the maximal retrieval capacity reachable by clamping.

Phase diagram ($T = 0$)

The $T = 0$ diagram in the (α, t) plane has three regions:

t	y^*	$\alpha_c(t)$
1.00	1.513	0.1379
0.95	1.47	0.1417
0.85	1.37	0.1515
$t^* \approx 0.713$	1.00	0.1753
$t < t^*$	—	spinodal absent

Table 1: Retrieval spinodal $\alpha_c(t)$ at $T = 0$ from Equation 42.

- **Retrieval region** ($t \geq t^*$ and $\alpha < \alpha_c(t)$, or $t < t^*$): the large- y root of Equation 36 gives a high- m solution that is locally stable. $\alpha_c(t)$ is a monotone decreasing function of t on $[t^*, 1]$ going from the maximal capacity ≈ 0.1753 at t^* down to the classical AGS value ≈ 0.138 at $t = 1$.

- **First-order loss of retrieval** ($t \in [t^*, 1]$, $\alpha > \alpha_c(t)$): the retrieval branch has annihilated with the intermediate unstable fixed point, leaving only a low- m solution. The jump in m across this line is the remnant of the classical AGS first-order transition.
- **No spinodal** ($t < t^*$): Equation 36 has a unique root at any $\alpha > 0$, and $m(\alpha; t)$ is a smooth, monotone-decreasing function of α that stays strictly positive. Clamping more than a fraction $1 - t^* \approx 0.287$ of the spins is enough to suppress the first-order retrieval transition altogether: the retrieval attractor persists continuously up to arbitrarily large α .

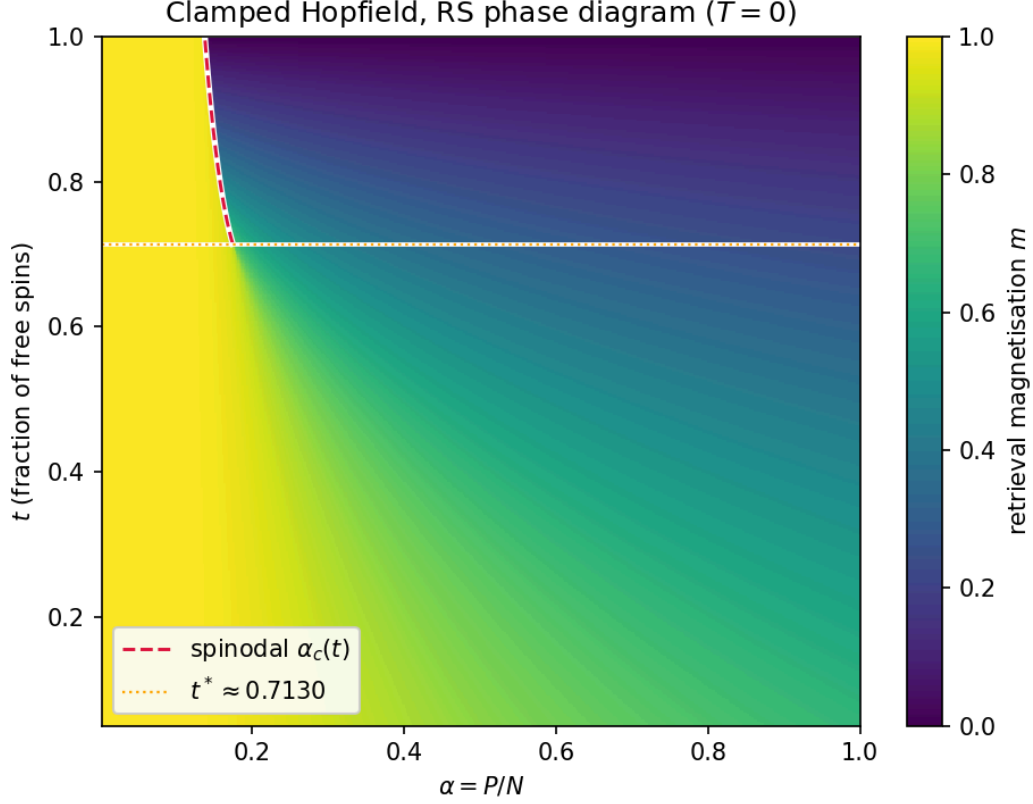


Figure 1: Retrieval magnetisation m at $T = 0$ in the (α, t) plane, from the largest-root solution of Equation 36. The red dashed curve is the spinodal $\alpha_c(t)$ from Equation 39, which terminates at $(\alpha, t) = (\approx 0.175, t^* \approx 0.713)$. Below t^* (dotted orange line) the transition is absent and m varies smoothly.

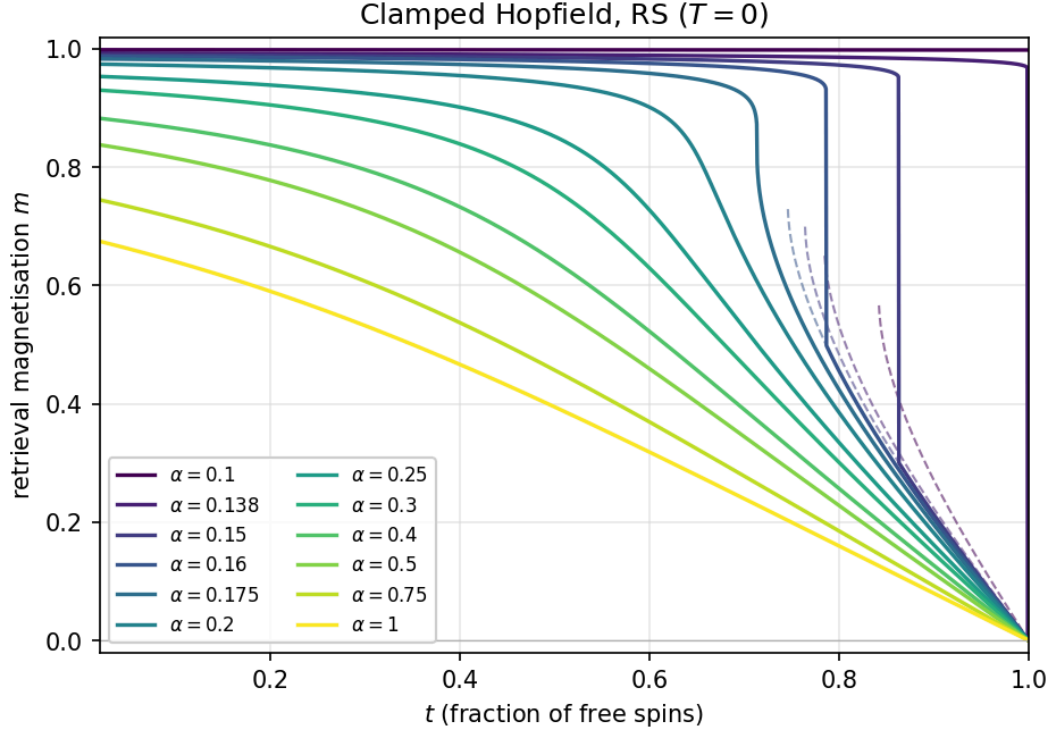


Figure 2: Retrieval magnetisation m versus t at $T = 0$ for several values of α . For $\alpha \leq \alpha_c(t^*) \approx 0.175$ (darker curves) the retrieval branch terminates at a first-order jump at some $t_c(\alpha)$; for $\alpha > \alpha_c(t^*)$ (lighter curves) $m(t)$ is smooth for all $t \in (0, 1]$. All curves approach $m = \text{erf}(1/\sqrt{2\alpha})$ as $t \rightarrow 0$.

MCMC validation at $T = 0.01$

The two figures below compare the $T = 0$ theory curves with MCMC runs at $\beta = 100$ ($T = 0.01$), a temperature low enough that finite- T corrections are negligible. Each dot is the average magnetisation over 10 independent chains after 400 sweeps on a system of $N = 20000$ spins.

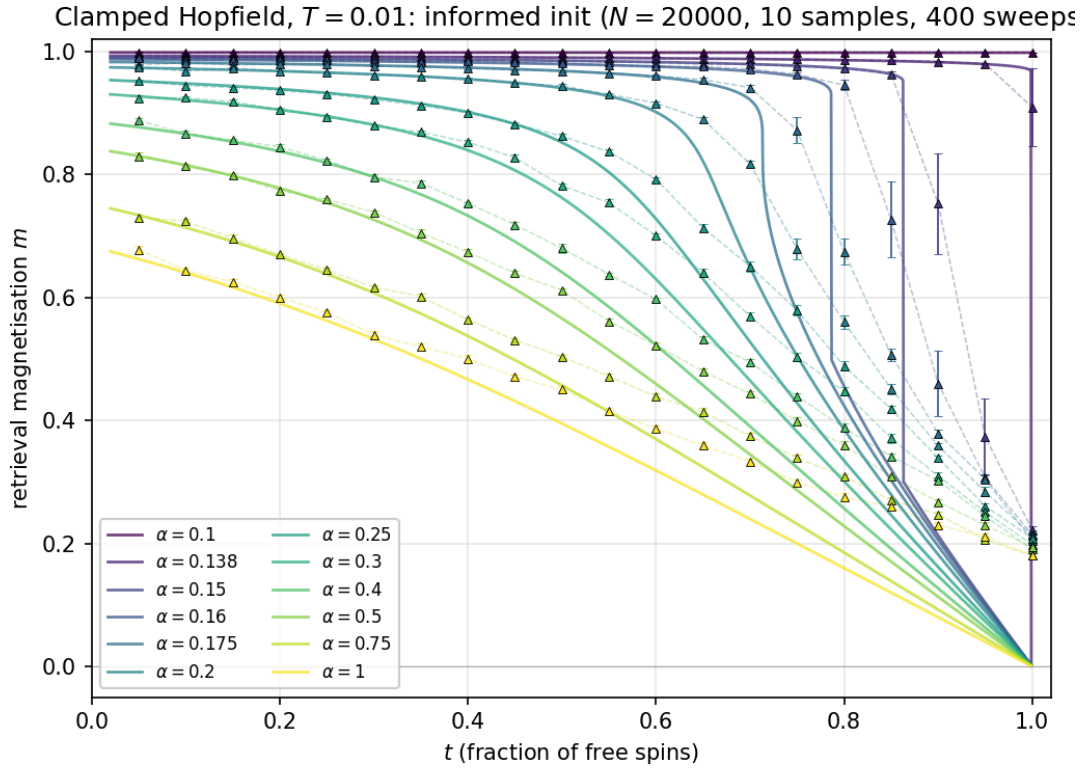


Figure 3: MCMC at $T = 0.01$, chains initialised on pattern 1 (informed initialisation). Solid curves are the $T = 0$ theory prediction. The pattern-started chains track the retrieval branch cleanly, confirming the spinodal boundary $\alpha_c(t)$.

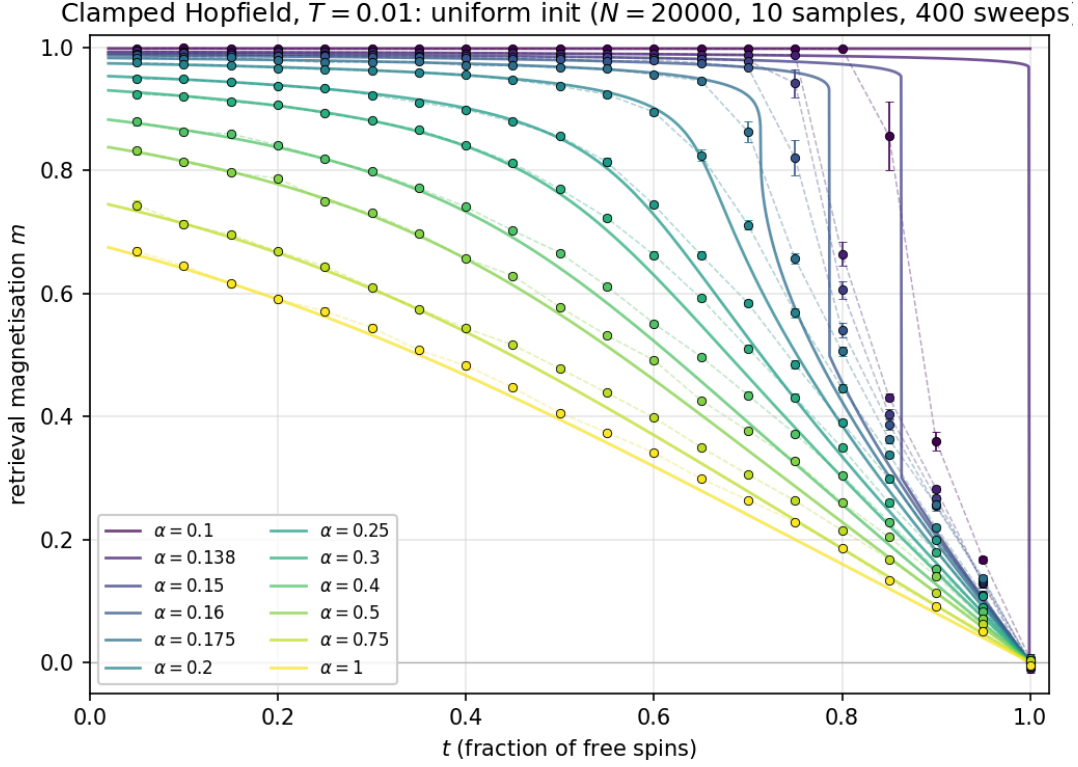


Figure 4: MCMC at $T = 0.01$, chains initialised uniformly at random. Solid curves are the $T = 0$ theory prediction. Random-started chains converge to the low- m (paramagnetic or spin-glass) branch and show a sharp jump onto the retrieval branch as t decreases below the spinodal, consistent with the first-order nature of the transition for $t \geq t^*$.

Remarks

- **Clamping as an external field.** The clamped block enters the effective local field on free sites through $M = (1 - t) + tm$, i.e. it contributes a constant bias $\beta(1 - t)$ irrespective of m . This single observation explains both (i) the upward shift of α_c as t decreases (more clamping = stronger field = larger retrieval basin), and (ii) the eventual disappearance of the first-order transition below t^* (the field becomes large enough to keep the retrieval fixed point isolated from any competing attractor at all α).
- **Global vs local transition.** Within RS, $\alpha_c(t)$ is the local stability (spinodal) boundary of the retrieval branch. A first-order thermodynamic transition, defined by equality of free energies between the retrieval and spin-glass branches, lies at a slightly smaller $\alpha_{th}(t)$. In the standard Hopfield model the gap $\alpha_c - \alpha_{th}$ is already very small ($\alpha_c \approx 0.138$ vs. $\alpha_{th} \approx 0.137$) and we expect the same to hold here.
- **Replica-symmetry breaking.** The de Almeida–Thouless (AT) instability of RS is controlled by the replicon eigenvalue of the free-energy Hessian. With the substitution $q \rightarrow \tilde{q}$ in the noise kernel, the AT condition takes the form

$$\alpha t^2 \int Dz \operatorname{sech}^4(\beta(M + \sqrt{\alpha r} z)) = (1 - \beta(1 - \tilde{q}))^2. \quad (43)$$

Its $T = 0$ limit should determine a narrow RSB pocket near the spinodal, qualitatively similar to the one in the classical Hopfield phase diagram.